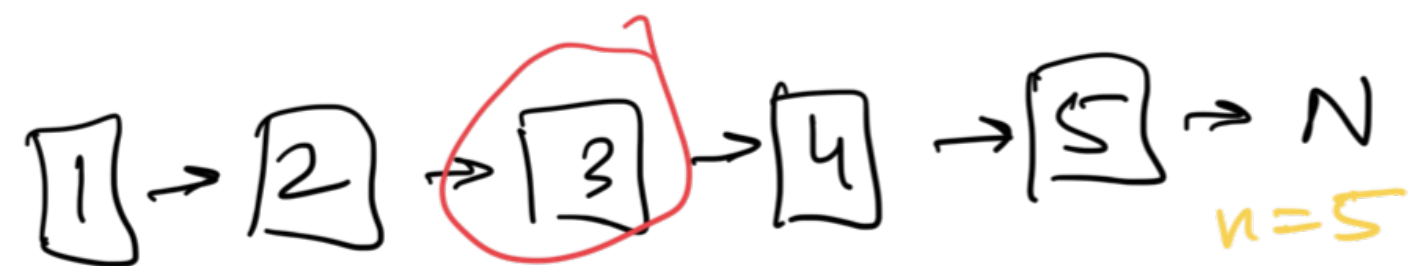
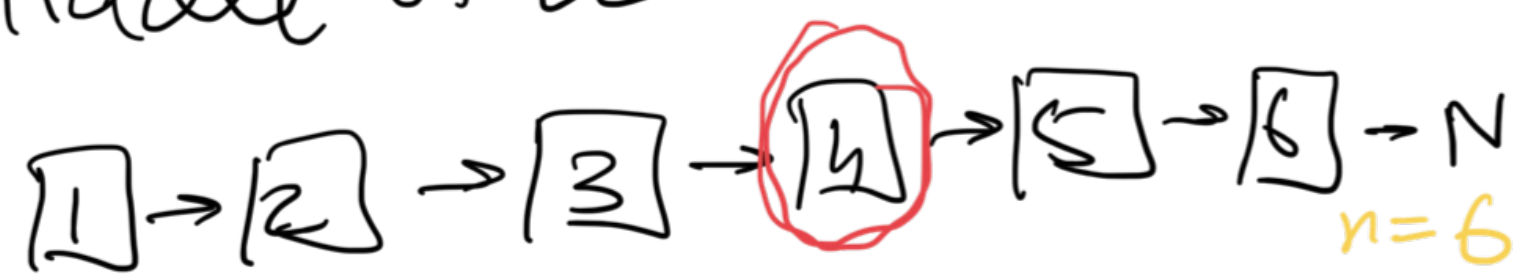


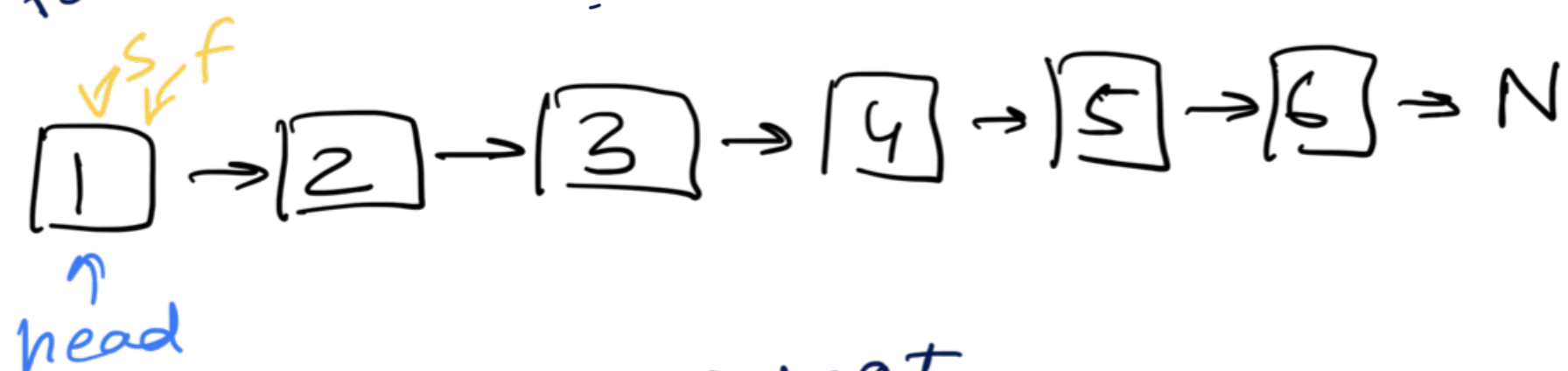
Middle of LL



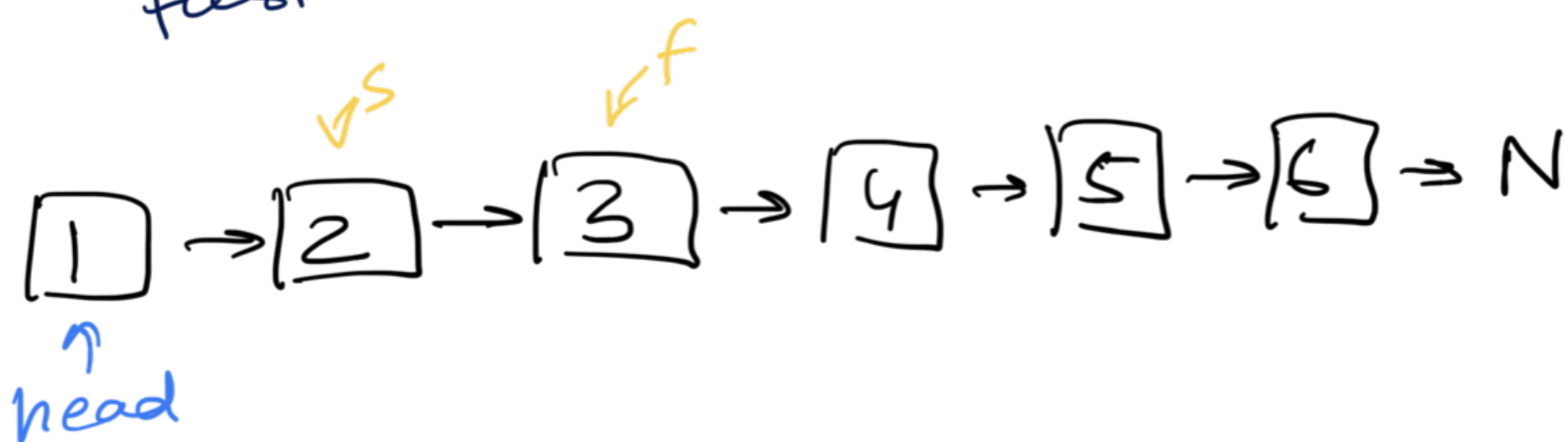
① calc size then
 $\left\lfloor \frac{\text{size}}{2} + 1 \right\rfloor^{\text{th}}$ node is middle

② slow - Fast approach
 → single loop

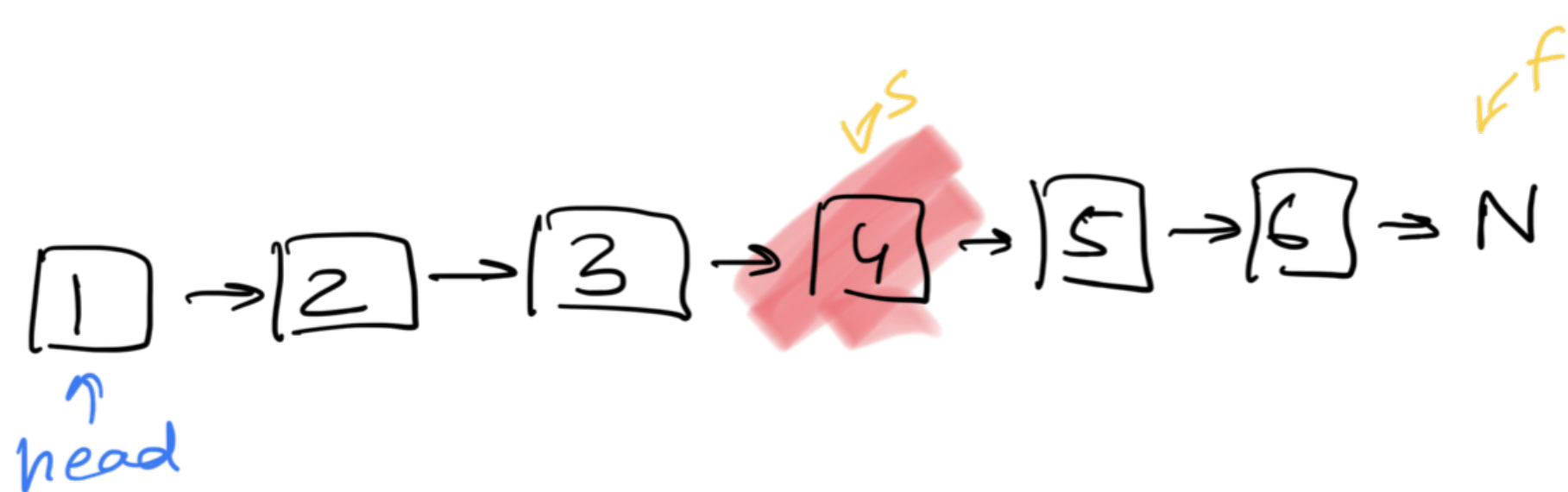
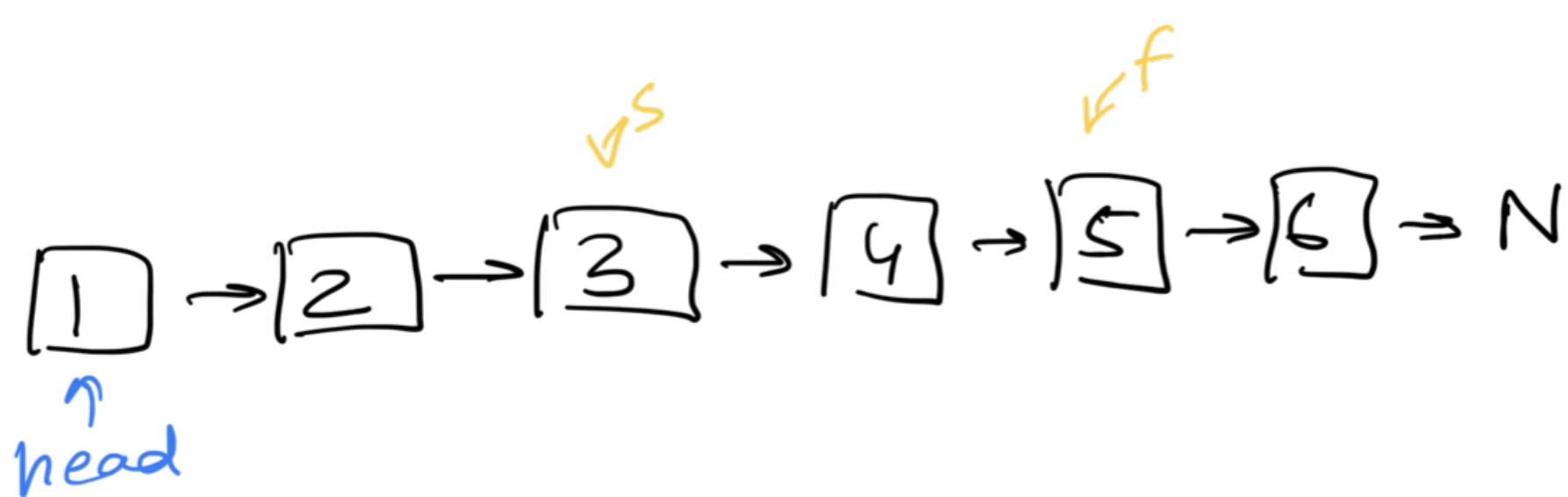
slow = head
 fast = head



slow = slow → next
 fast = fast → next → next



As soon as fast reaches the end the
 slow ptr will definitely point to
 the middle node ALWAYS



THE PROBLEM

in even sized stop when
 fast is pointing at null

in odd sized stop when
 fast → next is pointing at null

```

slow = head
fast = head
while (fast != NULL && fast->next != NULL) {
    slow = slow->next
    fast = fast->next->next
}
return slow;
    
```

SC: $O(1)$

TC: $O(n)$

